Git Lab Experiment 4 to 6

Program 4

Collaboration and Remote Repositories:

Clone a remote Git repository to your local machine.

To clone a remote Git repository to your local machine, you can use the git clone command. Here's the general syntax:

git clone<repository_url>

Replace <repository_url> with the actual URL of the Git repository you want to clone. For example:

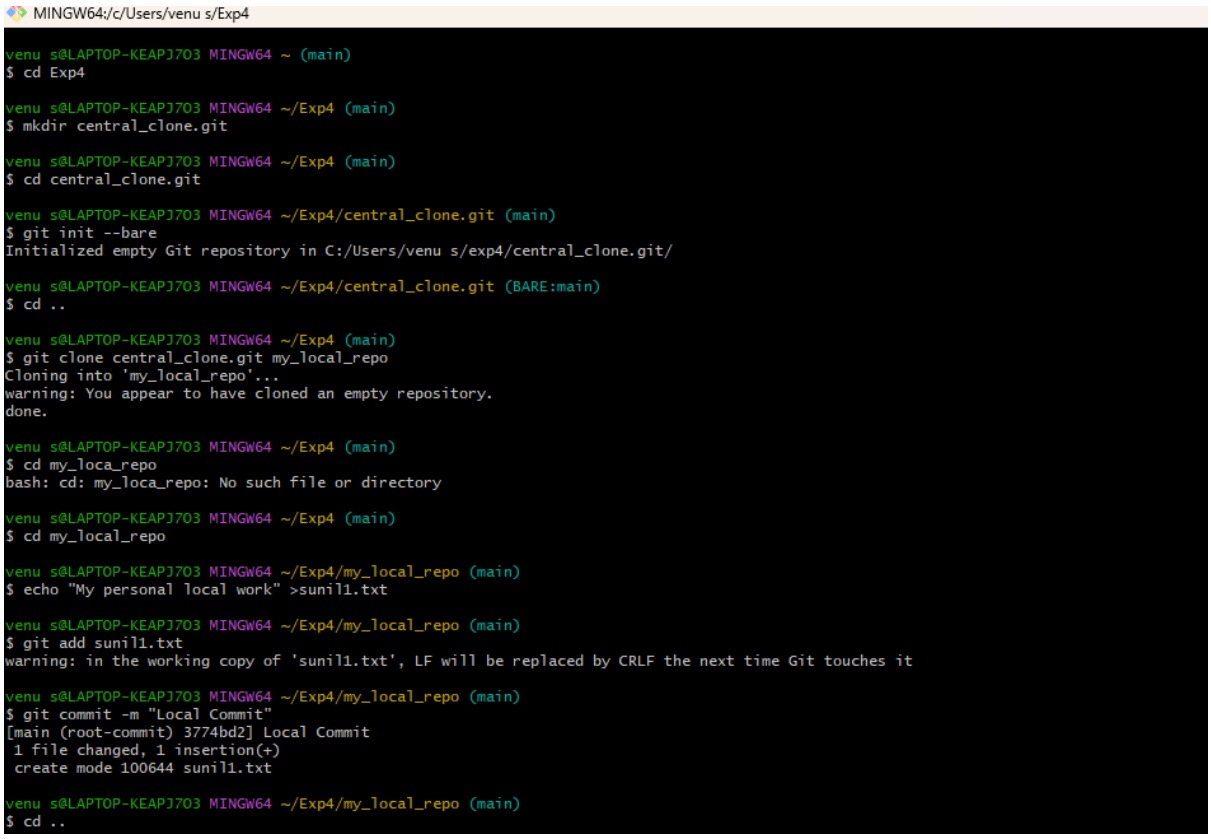
git clone https://github.com/example/repo.git

This command will create a new directory with the name of the repository and download all the files from the remote repository into that directory. If the repository is private and requires authentication, you might need to use the SSH URL or provide your credentials during the cloning process.

For SSH:

git clone git@github.com:example/repo.git

After running the git clone command, you'll have a local copy of the remote repository on your machine, and you can start working with the code.



```
venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/my_local_repo (main)
$ cd ..

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4 (main)
$ git clone central_clone.git teammate_clone
Cloning into 'teammate_clone'...
warning: You appear to have cloned an empty repository.
done.

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4 (main)
$ cd teammate_clone

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/teammate_clone (main)
$ git checkout -b master
Switched to a new branch 'master'

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/teammate_clone (master)
$ echo "Upstream commit" > sunil2.txt

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/teammate_clone (master)
$ git add .
warning: in the working copy of 'sunil2.txt', LF will be replaced by CRLF the next time Git touches it

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/teammate_clone (master)
$ git commit -m "Upstream commit"
[master (root-commit) bc7076e] Upstream commit
 1 file changed, 1 insertion(+)
 create mode 100644 sunil2.txt

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/teammate_clone (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 236 bytes | 39.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To C:/Users/venu s/exp4/central_clone.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

```
venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/teammate_clone (master)
$ cd ../my_local_repo

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/my_local_repo (main)
$ git fetch origin
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 216 bytes | 10.00 KiB/s, done.
From C:/Users/venu s/exp4/central_clone
 * [new branch]      master     -> origin/master

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/my_local_repo (main)
$ git log --oneline --all --graph
* bc7076e (origin/master) Upstream commit
* 3774bd2 (HEAD -> main) Local Commit

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp4/my_local_repo (main)
$
```

Program 5

Collaboration and Remote Repositories:

Fetch the latest changes from a remote repository and rebase your local branch onto the updated remote branch.

1. Fetch the latest changes from the remote repository:

git fetch

This command fetches the latest changes from the remote repository without automatically merging them into your local branches.

2. Rebase your local branch onto the updated remote branch:

Assuming you are currently on the branch you want to update (replace your-branch with the actual name of your branch):

git rebase origin/your-branch

This command applies your local commits on top of the changes fetched from the remote branch. If

conflicts arise, Git will pause the rebase process and ask you to resolve them. Alternatively, you can use

the interactive rebase to review and modify commits during the rebase:

git rebase -i origin/your-branch

This opens an editor where you can pick, squash, or edit individual commits.

3. Continue the rebase or resolve conflicts:

If conflicts occurred during the rebase, Git will prompt you to resolve them. After resolving conflicts, you can continue the rebase with:

git rebase –continue

If you decide to abort the rebase at any point, you can use:

git rebase –abort

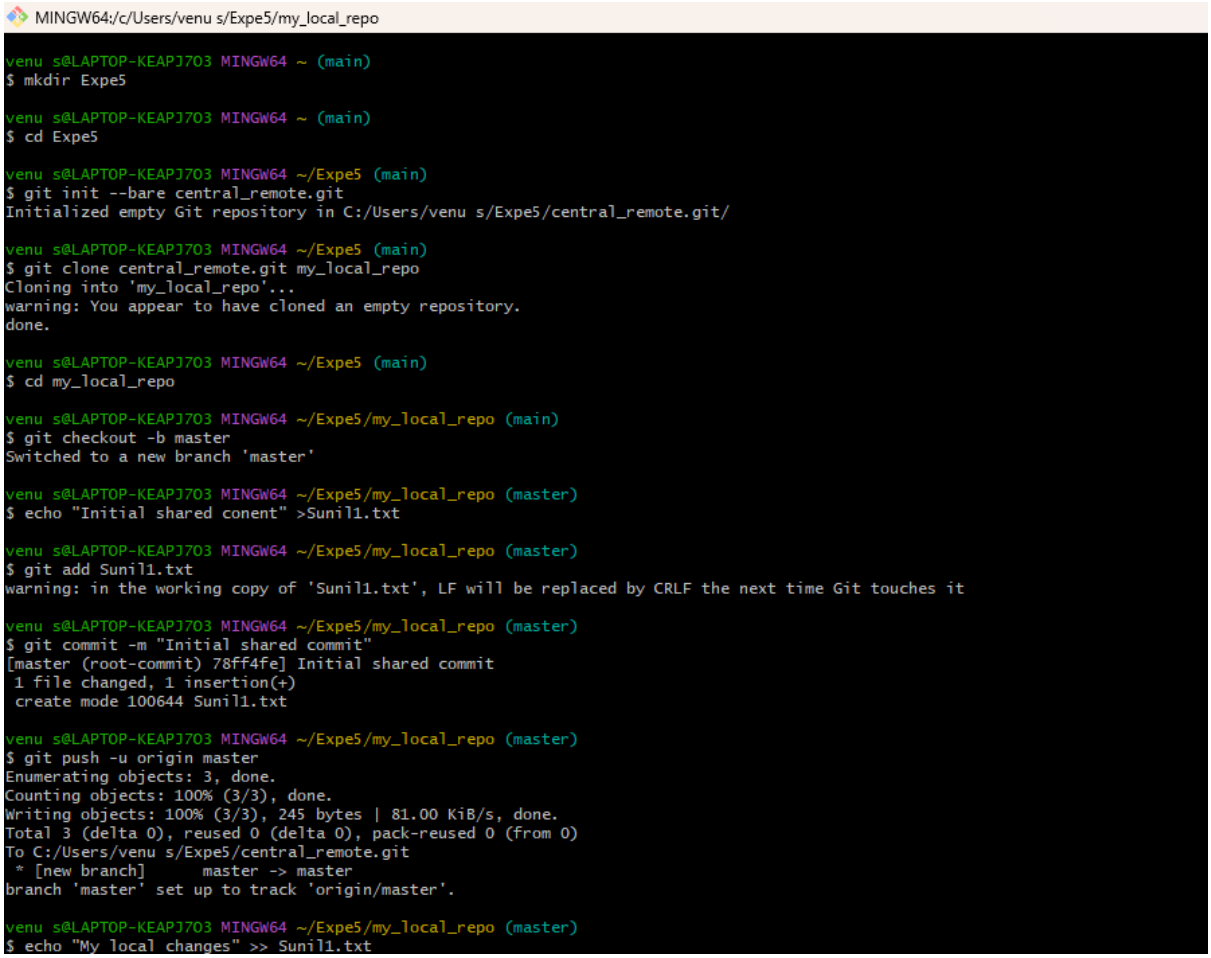
4. Push the rebased branch to the remote repository:

After successfully rebasing your local branch, you may need to force-push the changes to the remote

repository:

git push origin your-branch –force

Be cautious with force-pushing, especially if others are working with the same branch, as it rewrites the commit history. Now, your local branch is rebased onto the updated remote branch. Keep in mind that force-pushing should be done with caution, especially on shared branches, to avoid disrupting collaborative work.



```
venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/teammate_clone (main)
$ cd ..

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5 (main)
$ git clone central_remote.git teammate_clone
fatal: destination path 'teammate_clone' already exists and is not an empty directory.

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5 (main)
$ cd teammate_clone

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/teammate_clone (main)
$ git checkout master
branch 'master' set up to track 'origin/master'.
Switched to a new branch 'master'

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/teammate_clone (master)
$ echo "Teammate changes" >>Sunil1.txt

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/teammate_clone (master)
$ git add Sunil1.txt
warning: in the working copy of 'Sunil1.txt', LF will be replaced by CRLF the next time Git touches it

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/teammate_clone (master)
$ git commit -m "Add teammate changes"
[master 44efe76] Add teammate changes
 1 file changed, 1 insertion(+)

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/teammate_clone (master)
$ git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 293 bytes | 97.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To C:/Users/venu s/Expe5/central_remote.git
   78ff4fe..44efe76  master -> master

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/teammate_clone (master)
$ cd ../my_local_repo

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/my_local_repo (master)
$ git fetch origin
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 273 bytes | 10.00 KiB/s, done.
From C:/Users/venu s/Expe5/central_remote
   78ff4fe..44efe76  master     -> origin/master
```

```
venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/my_local_repo (master)
$ git rebase origin/master
Auto-merging Sunil1.txt
CONFLICT (content): Merge conflict in Sunil1.txt
error: could not apply b61bf0e... Add local Changes
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply b61bf0e... # Add local Changes

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/my_local_repo (master|REBASE 1/1)
$ git log --oneline  --graph
* 44efe76 (HEAD, origin/master) Add teammate changes
* 78ff4fe Initial shared commit

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/my_local_repo (master|REBASE 1/1)
$ git add Sunil1.txt

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/my_local_repo (master|REBASE 1/1)
$ git rebase --continue
[detached HEAD b9bf334] ;rgb:bfbf/bfbf/bfbf
 1 file changed, 4 insertions(+)
Successfully rebased and updated refs/heads/master.

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Expe5/my_local_repo (master)
$ git rebase --abort
fatal: no rebase in progress
```

Program 6

Collaboration and Remote Repositories:

Write the command to merge "feature-branch" into "master" while providing a custom commit message for the merge.

To merge “feature-branch” into “master” and provide a custom commit message, you can use the following command:

git merge feature-branch -m "Your custom commit message"

Replace “Your custom commit message” with the actual message you want to use for the merge commit. This command performs the merge and creates a new commit on the “master” branch with the specified message. If there are no conflicts, Git will complete the merge automatically.

If conflicts occur during the merge, Git will pause and prompt you to resolve the conflicts manually. After resolving conflicts, you can continue the merge process with:

git merge –continue

Alternatively, you can use an interactive merge to modify the commit message before finalizing the merge:

git merge feature-branch --no-ff –e

This opens an editor where you can edit the commit message before completing the merge. Again, replace “feature-branch” with the name of your actual feature branch.

MINGW64:/c/Users/venu s/Exp6

```
venu s@LAPTOP-KEAPJ7O3 MINGW64 ~ (main)
$ mkdir Exp6
mkdir: cannot create directory ‘Exp6’: File exists

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~ (main)
$ cd Exp6

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master)
$ git init
Reinitialized existing Git repository in C:/Users/venu s/Exp6/.git/

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master)
$ git checkout -b master
fatal: a branch named 'master' already exists

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master)
$ echo "Project baseline" > sunil.html

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master)
$ git add sunil.html
warning: in the working copy of 'sunil.html', LF will be replaced by CRLF the next time Git touches it

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master)
$ git commit -m "Initial commit"
[master 5411eea] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 sunil.html

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master)
$ git checkout -b feature_branch
fatal: a branch named 'feature_branch' already exists

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master)
$ git checkout  feature_branch
Switched to branch 'feature_branch'

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (feature_branch)
$ echo "Adding authentication module" >> sunil.html

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (feature_branch)
$ git add sunil.html
warning: in the working copy of 'sunil.html', LF will be replaced by CRLF the next time Git touches it

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (feature_branch)
$ git commit -m "Add authentication module"
[feature_branch 3360c56] Add authentication module
 1 file changed, 1 insertion(+)
 create mode 100644 sunil.html
```

```
venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (feature_branch)
$ git checkout master
Switched to branch 'master'

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master)
$ git merge feature_branch --no-ff -m "Integrated auth module"
Auto-merging sunil.html
CONFLICT (add/add): Merge conflict in sunil.html
Automatic merge failed; fix conflicts and then commit the result.

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master|MERGING)
$ git log --oneline --graph -n
error: -n requires an argument

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master|MERGING)
$ git log --oneline --graph -n 3
* 5411eea (HEAD -> master) Initial commit
*   8722021 Merge branch 'feature_branch' into master: Integrated auth module
|\
| * 9b23869 Feat: Add authentication module
|/

venu s@LAPTOP-KEAPJ7O3 MINGW64 ~/Exp6 (master|MERGING)
$
```